

## ANALISIS INSIGHT

### A. BUAT TABEL DAN IMPOR DATA

```
CREATE TABLE online_retail (  
    InvoiceNo TEXT,  
    StockCode TEXT,  
    Description TEXT,  
    Quantity INT,  
    InvoiceDate TEXT,  
    UnitPrice TEXT,  
    CustomerID INT,  
    Country TEXT  
);
```

Data Output Messages Notifications

CREATE TABLE

Query returned successfully in 104 msec.

**\*CATATAN** : Karna tipe data yang di buat di tabel online\_retail berbeda dengan punya meli jadi setiap perhitungan di berikan

query (quantity \* REPLACE(unitprice, ',', '.'):NUMERIC)

```
\copy online_retail FROM 'C:/Users/frisc/PROJEK MANDIRI/Online Retail 3.csv'  
DELIMITER ';' CSV HEADER;
```

```
Retail=# \copy online_retail FROM 'C:/Users/frisc/PROJEK MANDIRI/Online Retail 3.csv' DELIMITER ';' CSV HEADER;  
COPY 541909  
Retail=#
```

```
SELECT*FROM online_retail;
```

Data Output Messages Notifications

|    | invoiceno<br>text | stockcode<br>text | description<br>text                 | quantity<br>integer | invoicedate<br>text | unitprice<br>text | customerid<br>integer | country<br>text |
|----|-------------------|-------------------|-------------------------------------|---------------------|---------------------|-------------------|-----------------------|-----------------|
| 1  | 536365            | 85123A            | WHITE HANGING HEART T-LIGHT HOLDER  | 6                   | 01/12/10 08.26      | 2,55              | 17850                 | United Kingdom  |
| 2  | 536365            | 71053             | WHITE METAL LANTERN                 | 6                   | 01/12/10 08.26      | 3,39              | 17850                 | United Kingdom  |
| 3  | 536365            | 84406B            | CREAM CUPID HEARTS COAT HANGER      | 8                   | 01/12/10 08.26      | 2,75              | 17850                 | United Kingdom  |
| 4  | 536365            | 84029G            | KNITTED UNION FLAG HOT WATER BOTTLE | 6                   | 01/12/10 08.26      | 3,39              | 17850                 | United Kingdom  |
| 5  | 536365            | 84029E            | RED WOOLLY HOTTIE WHITE HEART.      | 6                   | 01/12/10 08.26      | 3,39              | 17850                 | United Kingdom  |
| 6  | 536365            | 22752             | SET 7 BABUSHKA NESTING BOXES        | 2                   | 01/12/10 08.26      | 7,65              | 17850                 | United Kingdom  |
| 7  | 536365            | 21730             | GLASS STAR FROSTED T-LIGHT HOLDER   | 6                   | 01/12/10 08.26      | 4,25              | 17850                 | United Kingdom  |
| 8  | 536366            | 22633             | HAND WARMER UNION JACK              | 6                   | 01/12/10 08.28      | 1,85              | 17850                 | United Kingdom  |
| 9  | 536366            | 22632             | HAND WARMER RED POLKA DOT           | 6                   | 01/12/10 08.28      | 1,85              | 17850                 | United Kingdom  |
| 10 | 536367            | 84879             | ASSORTED COLOUR BIRD ORNAMENT       | 32                  | 01/12/10 08.34      | 1,69              | 13047                 | United Kingdom  |

```
ALTER TABLE online_retail  
ADD COLUMN kategori_utama TEXT;
```

Data Output Messages Notifications

ALTER TABLE

Query returned successfully in 109 msec.

## B. ANALISIS REVENUE

### 1. NET REVENUE

```
SELECT
SUM(CASE
    WHEN REPLACE(unitprice, ',', '. '::NUMERIC > 0
    THEN quantity * REPLACE(unitprice, ',', '. '::NUMERIC
    ELSE 0
END) AS total_revenue_positif,
SUM(CASE
    WHEN REPLACE(unitprice, ',', '. '::NUMERIC < 0
    THEN quantity * REPLACE(unitprice, ',', '. '::NUMERIC
    ELSE 0
END) AS total_return,
SUM(quantity * REPLACE(unitprice, ',', '. '::NUMERIC) AS revenue_bersih
FROM online_retail;
```

| Data Output Messages Notifications |                                  |                         |                           |
|------------------------------------|----------------------------------|-------------------------|---------------------------|
|                                    | total_revenue_positif<br>numeric | total_return<br>numeric | revenue_bersih<br>numeric |
| 1                                  | 9769872.054                      | -22124.12               | 9747747.934               |

### 2. Total transaksi dan revenue per kategori

```
UPDATE online_retail
SET kategori_utama = CASE
    WHEN stockcode IN ('AMAZONFEE', 'BANK CHARGES', 'D', 'DOT', 'POST',
        'PADS', 'CRUK', 'C2', 'B', 'M', 'S', 'gift_000')
    THEN 'Non-Produk'
    ELSE 'Produk Asli'
END;
SELECT kategori_utama,
COUNT(*) AS total_transaksi,
SUM(quantity * REPLACE(unitprice, ',', '. '::NUMERIC) AS total_revenue
FROM online_retail
WHERE REPLACE(unitprice, ',', '. '::NUMERIC > 0
GROUP BY kategori_utama;
```

| Data Output Messages Notifications |                        |                           |                          |
|------------------------------------|------------------------|---------------------------|--------------------------|
|                                    | kategori_utama<br>text | total_transaksi<br>bigint | total_revenue<br>numeric |
| 1                                  | Non-Produk             | 2913                      | -23525.186               |
| 2                                  | Produk Asli            | 538994                    | 9793397.24               |

### 3. Total transaksi dan revenue per Negara

```
SELECT country,  
       COUNT(*) AS total_transaksi,  
       SUM(quantity::NUMERIC * REPLACE(unitprice, ',', ':')::NUMERIC) AS  
total_revenue  
FROM online_retail  
GROUP BY country  
ORDER BY total_revenue DESC;
```

| Data Output Messages Notifications         |                 |                           |                          |
|--------------------------------------------|-----------------|---------------------------|--------------------------|
|                                            | country<br>text | total_transaksi<br>bigint | total_revenue<br>numeric |
| 1                                          | United Kingdom  | 495478                    | 8187806.364              |
| 2                                          | Netherlands     | 2371                      | 284661.54                |
| 3                                          | EIRE            | 8196                      | 263276.82                |
| 4                                          | Germany         | 9495                      | 221698.21                |
| 5                                          | France          | 8557                      | 197403.90                |
| 6                                          | Australia       | 1259                      | 137077.27                |
| 7                                          | Switzerland     | 2002                      | 56385.35                 |
| 8                                          | Spain           | 2533                      | 54774.58                 |
| 9                                          | Belgium         | 2069                      | 40910.96                 |
| 10                                         | Sweden          | 462                       | 36595.91                 |
| 11                                         | Japan           | 358                       | 35340.62                 |
| Total rows: 38 Query complete 00:00:00.538 |                 |                           |                          |

### 4. Negara dengan jumlah transaksi terbanyak

```
SELECT country,  
       COUNT(*) AS total_transaksi  
FROM online_retail  
GROUP BY country  
ORDER BY total_transaksi DESC  
LIMIT 1;
```

| Data Output Messages Notifications |                 |                           |
|------------------------------------|-----------------|---------------------------|
|                                    | country<br>text | total_transaksi<br>bigint |
| 1                                  | United Kingdom  | 495478                    |

### 5. Total Revenue/Bulan

```
SELECT DATE_TRUNC('month', TO_TIMESTAMP(invoicedate, 'DD/MM/YY  
HH24:MI')) AS bulan,  
       SUM(quantity::NUMERIC * REPLACE(unitprice, ',', ':')::NUMERIC) AS  
total_revenue  
FROM online_retail  
GROUP BY bulan  
ORDER BY bulan;
```

| Data Output Messages Notifications |                                   |                          |
|------------------------------------|-----------------------------------|--------------------------|
|                                    | bulan<br>timestamp with time zone | total_revenue<br>numeric |
| 1                                  | 2010-12-01 00:00:00+07            | 748957.02                |
| 2                                  | 2011-01-01 00:00:00+07            | 560000.26                |
| 3                                  | 2011-02-01 00:00:00+07            | 498062.65                |
| 4                                  | 2011-03-01 00:00:00+07            | 683267.08                |
| 5                                  | 2011-04-01 00:00:00+07            | 493207.121               |
| 6                                  | 2011-05-01 00:00:00+07            | 723333.51                |
| 7                                  | 2011-06-01 00:00:00+07            | 691123.12                |
| 8                                  | 2011-07-01 00:00:00+07            | 681300.111               |
| 9                                  | 2011-08-01 00:00:00+07            | 682680.51                |
| 10                                 | 2011-09-01 00:00:00+07            | 1019687.622              |
| 11                                 | 2011-10-01 00:00:00+07            | 1070704.67               |
| 12                                 | 2011-11-01 00:00:00+07            | 1461756.25               |
| 13                                 | 2011-12-01 00:00:00+07            | 433668.01                |

## 6. Jumlah Revenue per Minggu

### Hari

```
SELECT TO_CHAR(TO_TIMESTAMP(invoicedate, 'DD/MM/YY HH24.MI'),
'Day') AS hari,
COUNT(*) AS total_transaksi
FROM online_retail
GROUP BY hari
ORDER BY total_transaksi DESC;
```

| Data Output Messages Notifications |              |                           |
|------------------------------------|--------------|---------------------------|
|                                    | hari<br>text | total_transaksi<br>bigint |
| 1                                  | Thursday     | 103857                    |
| 2                                  | Tuesday      | 101808                    |
| 3                                  | Monday       | 95111                     |
| 4                                  | Wednesday    | 94565                     |
| 5                                  | Friday       | 82193                     |
| 6                                  | Sunday       | 64375                     |

### Minggu

```
SELECT TO_CHAR(TO_TIMESTAMP(invoicedate, 'DD/MM/YY HH24.MI'),
'Week') AS Minggu,
COUNT(*) AS total_transaksi
FROM online_retail
GROUP BY Minggu
ORDER BY total_transaksi DESC;
```

| Data Output Messages Notificati |                |                           |
|---------------------------------|----------------|---------------------------|
|                                 | minggu<br>text | total_transaksi<br>bigint |
| 1                               | 1eek           | 137656                    |
| 2                               | 2eek           | 131658                    |
| 3                               | 3eek           | 124825                    |
| 4                               | 4eek           | 109912                    |
| 5                               | 5eek           | 37858                     |

## Bulan

```
SELECT TO_CHAR(TO_TIMESTAMP(invoicedate, 'DD/MM/YY HH24.MI'),
'Month') AS Bulan,
COUNT(*) AS total_transaksi
FROM online_retail
GROUP BY Bulan
ORDER BY total_transaksi DESC;
```

| Data Output Messages Notifications |               |                           |
|------------------------------------|---------------|---------------------------|
|                                    | bulan<br>text | total_transaksi<br>bigint |
| 1                                  | November      | 84711                     |
| 2                                  | December      | 68006                     |
| 3                                  | October       | 60742                     |
| 4                                  | September     | 50226                     |
| 5                                  | July          | 39518                     |
| 6                                  | May           | 37030                     |
| 7                                  | June          | 36874                     |
| 8                                  | March         | 36748                     |
| 9                                  | August        | 35284                     |
| 10                                 | January       | 35147                     |
| 11                                 | April         | 29916                     |
| 12                                 | February      | 27707                     |

## 7. Top Revenue Customer

```
SELECT customerid,
SUM(quantity::NUMERIC * REPLACE(unitprice, ',', '::')::NUMERIC) AS
total_revenue
FROM online_retail
GROUP BY customerid
ORDER BY total_revenue DESC;
```

| Data Output Messages Notifications |                       |                          |
|------------------------------------|-----------------------|--------------------------|
|                                    | customerid<br>integer | total_revenue<br>numeric |
| 1                                  | [null]                | 1447682.12               |
| 2                                  | 14646                 | 279489.02                |
| 3                                  | 18102                 | 256438.49                |
| 4                                  | 17450                 | 187482.17                |
| 5                                  | 14911                 | 132572.62                |
| 6                                  | 12415                 | 123725.45                |
| 7                                  | 14156                 | 113384.14                |
| 8                                  | 17511                 | 88125.38                 |
| 9                                  | 16684                 | 65892.08                 |
| 10                                 | 13694                 | 62653.10                 |
| 11                                 | 15311                 | 59419.34                 |
| 12                                 | 13089                 | 57385.88                 |
| 13                                 | 14096                 | 57120.91                 |
| Total rows: 4373                   |                       | Query complete 00:00:00  |

## 8. Top 5 negara dengan revenue terbesar by Produk

```

SELECT country,
       SUM(quantity::NUMERIC * REPLACE(unitprice, ',', '. '::NUMERIC)AS
total_revenue
FROM online_retail
WHERE kategori_utama = 'Produk Asli'
GROUP BY country
ORDER BY total_revenue DESC;

```

| Data Output Messages Notifications |                 |                           |
|------------------------------------|-----------------|---------------------------|
|                                    | country<br>text | total_revenue<br>numeric  |
| 1                                  | United Kingdom  | 8302406.64                |
| 2                                  | Netherlands     | 283479.54                 |
| 3                                  | EIRE            | 259663.46                 |
| 4                                  | Germany         | 200808.40                 |
| 5                                  | France          | 182163.39                 |
| 6                                  | Australia       | 136990.00                 |
| 7                                  | Switzerland     | 52505.35                  |
| 8                                  | Spain           | 51765.20                  |
| 9                                  | Belgium         | 36662.96                  |
| 10                                 | Japan           | 35419.79                  |
| 11                                 | Sweden          | 35176.91                  |
| 12                                 | Norway          | 32292.96                  |
| 13                                 | Portugal        | 26871.52                  |
| Total rows: 38                     |                 | Query complete 00:00:00.6 |

## 9. Rata2 transaksi customer

```

SELECT AVG(transaksi_per_customer) AS avg_transaksi
FROM (
  SELECT customerid, COUNT(*) AS transaksi_per_customer
  FROM online_retail
  GROUP BY customerid
) AS sub;

```

| Data Output |                      | Messages | Noti |
|-------------|----------------------|----------|------|
|             |                      |          |      |
|             | avg_transaksi        |          |      |
|             | numeric              |          |      |
| 1           | 123.9215641436085067 |          |      |

## 10. Rata2 penjualan quanti

```
SELECT AVG(quantity) AS avg_quantity
FROM online_retail;
```

| Data Output |                    | Messages | N |
|-------------|--------------------|----------|---|
|             |                    |          |   |
|             | avg_quantity       |          |   |
|             | numeric            |          |   |
| 1           | 9.5522495474332406 |          |   |

## 11. Top 10 Produk Paling Menguntungkan (Berdasarkan Profit Positif)

```
SELECT description,
       SUM(quantity::NUMERIC * REPLACE(unitprice, ',', '::NUMERIC) AS
total_profit
FROM online_retail
GROUP BY description
HAVING SUM(quantity::NUMERIC * REPLACE(unitprice, ',', '::NUMERIC) > 0
ORDER BY total_profit DESC
LIMIT 10;
```

| Data Output |                                    | Messages | Notifications |
|-------------|------------------------------------|----------|---------------|
|             |                                    |          |               |
|             | description                        |          |               |
|             | text                               |          |               |
|             | total_profit                       |          |               |
|             | numeric                            |          |               |
| 1           | DOTCOM POSTAGE                     |          | 206245.48     |
| 2           | REGENCY CAKESTAND 3 TIER           |          | 164762.19     |
| 3           | WHITE HANGING HEART T-LIGHT HOLDER |          | 99668.47      |
| 4           | PARTY BUNTING                      |          | 98302.98      |
| 5           | JUMBO BAG RED RETROSPOT            |          | 92356.03      |
| 6           | RABBIT NIGHT LIGHT                 |          | 66756.59      |
| 7           | POSTAGE                            |          | 66230.64      |
| 8           | PAPER CHAIN KIT 50'S CHRISTMAS     |          | 63791.94      |
| 9           | ASSORTED COLOUR BIRD ORNAMENT      |          | 58959.73      |
| 10          | CHILLI LIGHTS                      |          | 53768.06      |

## 12. Top 10 Produk yang merugikan

```
SELECT description,  
       SUM(quantity::NUMERIC * REPLACE(unitprice, ',', '.')::NUMERIC) AS  
kerugian  
FROM online_retail  
GROUP BY description  
HAVING SUM(quantity::NUMERIC * REPLACE(unitprice, ',', '.')::NUMERIC) < 0  
ORDER BY kerugian ASC  
LIMIT 10;
```

| Data Output Messages Notifications |                                |                     |
|------------------------------------|--------------------------------|---------------------|
|                                    | description<br>text            | kerugian<br>numeric |
| 1                                  | AMAZON FEE                     | -221520.50          |
| 2                                  | Manual                         | -68671.64           |
| 3                                  | Adjust bad debt                | -11062.06           |
| 4                                  | CRUK Commission                | -7933.43            |
| 5                                  | Bank Charges                   | -7175.639           |
| 6                                  | Discount                       | -5696.22            |
| 7                                  | SAMPLES                        | -3049.39            |
| 8                                  | WHITE CHERRY LIGHTS            | -54.00              |
| 9                                  | CREAM SWEETHEART MAGAZINE RACK | -46.85              |
| 10                                 | WOODEN BOX ADVENT CALENDAR     | -45.70              |

## C. ANALISIS PREDIKSI BULANAN

### 1. Transaksi Tanpa CID/Anonim

```
SELECT  
  
COUNT(*) AS transaksi_anonim,  
  
SUM(quantity::NUMERIC * REPLACE(unitprice, ',', '.')::NUMERIC) AS  
nilai_transaksi_anonim  
  
FROM online_retail  
  
WHERE customerid IS NULL;
```

| Data Output Messages Notifications |                            |                                   |
|------------------------------------|----------------------------|-----------------------------------|
|                                    | transaksi_anonim<br>bigint | nilai_transaksi_anonim<br>numeric |
| 1                                  | 135080                     | 1447682.12                        |

### 2. Tambahkan Kolom Angka Bulan untuk Linear Regression Manual

```
WITH penjualan_bulanan AS (  
SELECT  
DATE_TRUNC('month', TO_TIMESTAMP(invoicedate, 'DD/MM/YY HH24:MI')) AS  
bulan,
```



```

SUM(quantity::NUMERIC * REPLACE(unitprice, ',', '::')::NUMERIC) AS
total_penjualan
FROM online_retail
WHERE quantity::NUMERIC > 0 AND REPLACE(unitprice, ',', '::')::NUMERIC > 0
GROUP BY 1
),
data_berlabel AS (
SELECT
ROW_NUMBER() OVER (ORDER BY bulan) AS bulan_ke,
bulan,
total_penjualan
FROM penjualan_bulanan
)
SELECT * FROM data_berlabel;

```

| Data Output Messages Notifications                                                                                                                                                                                                                            |                    |                                   |                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-----------------------------------|----------------------------|
| <div> <div> <div>+</div> <div>📄</div> <div>▼</div> <div>📋</div> <div>▼</div> <div>🗑️</div> <div>📥</div> <div>⬇️</div> <div>📈</div> <div>SQL</div> </div> <div>Showing rows: 1 to 13 <span>✎</span> Page No: 1 of 1 <span>⏪</span> <span>⏩</span></div> </div> |                    |                                   |                            |
|                                                                                                                                                                                                                                                               | bulan_ke<br>bigint | bulan<br>timestamp with time zone | total_penjualan<br>numeric |
| 1                                                                                                                                                                                                                                                             | 1                  | 2010-12-01 00:00:00+07            | 823746.14                  |
| 2                                                                                                                                                                                                                                                             | 2                  | 2011-01-01 00:00:00+07            | 691364.56                  |
| 3                                                                                                                                                                                                                                                             | 3                  | 2011-02-01 00:00:00+07            | 523631.89                  |
| 4                                                                                                                                                                                                                                                             | 4                  | 2011-03-01 00:00:00+07            | 717639.36                  |
| 5                                                                                                                                                                                                                                                             | 5                  | 2011-04-01 00:00:00+07            | 537808.621                 |
| 6                                                                                                                                                                                                                                                             | 6                  | 2011-05-01 00:00:00+07            | 770536.02                  |
| 7                                                                                                                                                                                                                                                             | 7                  | 2011-06-01 00:00:00+07            | 761739.90                  |
| 8                                                                                                                                                                                                                                                             | 8                  | 2011-07-01 00:00:00+07            | 719221.191                 |
| 9                                                                                                                                                                                                                                                             | 9                  | 2011-08-01 00:00:00+07            | 759138.38                  |
| 10                                                                                                                                                                                                                                                            | 10                 | 2011-09-01 00:00:00+07            | 1058590.172                |
| 11                                                                                                                                                                                                                                                            | 11                 | 2011-10-01 00:00:00+07            | 1154979.30                 |
| 12                                                                                                                                                                                                                                                            | 12                 | 2011-11-01 00:00:00+07            | 1509496.33                 |
| 13                                                                                                                                                                                                                                                            | 13                 | 2011-12-01 00:00:00+07            | 638792.68                  |

### 3. Hitung Slope dan Intercept (Trendline Regression)

```

WITH data_berlabel AS (
SELECT
ROW_NUMBER() OVER (ORDER BY DATE_TRUNC('month',
TO_TIMESTAMP(invoicedate, 'DD/MM/YY HH24:MI')))) AS bulan_ke,
DATE_TRUNC('month', TO_TIMESTAMP(invoicedate, 'DD/MM/YY HH24:MI')) AS
bulan,
SUM(quantity::NUMERIC * REPLACE(unitprice, ',', '::')::NUMERIC) AS
total_penjualan
FROM online_retail
WHERE quantity::NUMERIC > 0 AND REPLACE(unitprice, ',', '::')::NUMERIC > 0
GROUP BY DATE_TRUNC('month', TO_TIMESTAMP(invoicedate, 'DD/MM/YY
HH24:MI'))
),
statistik AS (
SELECT

```

```

COUNT(*) AS n,
SUM(bulan_ke) AS sum_x,
SUM(total_penjualan) AS sum_y,
SUM(bulan_ke * total_penjualan) AS sum_xy,
SUM(bulan_ke * bulan_ke) AS sum_xx
FROM data_berlabel
),
regresi AS (
  SELECT
    (n * sum_xy - sum_x * sum_y) * 1.0 / (n * sum_xx - sum_x * sum_x) AS slope,
    (sum_y - ((n * sum_xy - sum_x * sum_y) * 1.0 / (n * sum_xx - sum_x * sum_x)) *
sum_x) / n AS intercept
  FROM statistik
)
SELECT
  d.bulan_ke,
  d.bulan,
  d.total_penjualan,
  ROUND(r.slope * (d.bulan_ke + 1) + r.intercept, 2) AS prediksi_bulan_depan,
  ROUND(r.slope * (d.bulan_ke + 2) + r.intercept, 2) AS prediksi_2_bulan_lagi
FROM data_berlabel d
CROSS JOIN regresi r
ORDER BY d.bulan_ke DESC
LIMIT 1;

```

| Data Output Messages Notifications   |                    |                                   |                            |                                 |                                  |
|--------------------------------------|--------------------|-----------------------------------|----------------------------|---------------------------------|----------------------------------|
| Showing rows: 1 to 1 Page No: 1 of 1 |                    |                                   |                            |                                 |                                  |
|                                      | bulan_ke<br>bigint | bulan<br>timestamp with time zone | total_penjualan<br>numeric | prediksi_bulan_depan<br>numeric | prediksi_2_bulan_lagi<br>numeric |
| 1                                    | 13                 | 2011-12-01 00:00:00+07            | 638792.68                  | 1086688.23                      | 1124713.09                       |

### Hasil:

- Dapat prediksi bulan depan dan dua bulan ke depan berdasarkan linear trend dari historis penjualan.
- Tidak sepresisi machine learning, tapi cukup kuat sebagai dasar manajemen inventory dan perencanaan.

| Temuan         | Implikasi Bisnis                                |
|----------------|-------------------------------------------------|
| Tren meningkat | Siapkan stok tambahan, perkuat kampanye iklan   |
| Tren menurun   | Evaluasi harga/promosi, aktifkan pelanggan lama |
| Tren stagnan   | Eksplorasi diversifikasi produk atau channel    |

## D. ANALISIS KOMBINASI PRODUK

### 1. Analisis Kombinasi Produk 2-Itemset

```

WITH transaksi_produk AS (
  SELECT

```

```

    invoiceno,
    LOWER(TRIM(description)) AS produk
FROM online_retail
WHERE quantity::NUMERIC > 0 AND REPLACE(unitprice, ',', '::NUMERIC > 0
    AND description IS NOT NULL
),
kombinasi_produk AS (
    SELECT
        a.produk AS produk_a,
        b.produk AS produk_b,
        COUNT(DISTINCT a.invoiceno) AS jumlah_kombinasi
    FROM transaksi_produk a
    JOIN transaksi_produk b
        ON a.invoiceno = b.invoiceno
        AND a.produk < b.produk -- Hindari duplikat dan self-join
    GROUP BY a.produk, b.produk
),
total_transaksi AS (
    SELECT COUNT(DISTINCT invoiceno) AS total_transaksi
    FROM transaksi_produk
)
SELECT
    k.produk_a,
    k.produk_b,
    k.jumlah_kombinasi,
    ROUND((k.jumlah_kombinasi * 100.0 / t.total_transaksi), 2) AS support_persen
FROM kombinasi_produk k
CROSS JOIN total_transaksi t
WHERE k.jumlah_kombinasi >= 30 -- minimal support threshold
ORDER BY k.jumlah_kombinasi DESC
LIMIT 20;

```

| Data Output Messages Notifications |                                   |                                    |                            |                           |  |
|------------------------------------|-----------------------------------|------------------------------------|----------------------------|---------------------------|--|
| Showing rows: 1                    |                                   |                                    |                            |                           |  |
|                                    | produk_a<br>text                  | produk_b<br>text                   | jumlah_kombinasi<br>bigint | support_persen<br>numeric |  |
| 1                                  | jumbo bag pink polkadot           | jumbo bag red retrospot            | 825                        | 4.13                      |  |
| 2                                  | green regency teacup and saucer   | roses regency teacup and saucer    | 767                        | 3.84                      |  |
| 3                                  | jumbo bag red retrospot           | jumbo storage bag suki             | 724                        | 3.63                      |  |
| 4                                  | jumbo bag red retrospot           | jumbo shopper vintage red paisl... | 680                        | 3.41                      |  |
| 5                                  | lunch bag black skull.            | lunch bag red retrospot            | 641                        | 3.21                      |  |
| 6                                  | alarm clock bakelike green        | alarm clock bakelike red           | 640                        | 3.21                      |  |
| 7                                  | green regency teacup and saucer   | pink regency teacup and saucer     | 632                        | 3.17                      |  |
| 8                                  | lunch bag pink polkadot           | lunch bag red retrospot            | 606                        | 3.04                      |  |
| 9                                  | pink regency teacup and saucer    | roses regency teacup and saucer    | 598                        | 3.00                      |  |
| 10                                 | jumbo bag baroque black white     | jumbo bag red retrospot            | 585                        | 2.93                      |  |
| 11                                 | jumbo bag red retrospot           | lunch bag red retrospot            | 579                        | 2.90                      |  |
| 12                                 | lunch bag red retrospot           | lunch bag spaceboy design          | 560                        | 2.81                      |  |
| 13                                 | lunch bag red retrospot           | lunch bag suki design              | 556                        | 2.79                      |  |
| 14                                 | lunch bag cars blue               | lunch bag red retrospot            | 554                        | 2.78                      |  |
| 15                                 | paper chain kit 50's christmas    | paper chain kit vintage christmas  | 552                        | 2.77                      |  |
| 16                                 | gardeners kneeling pad cup of tea | gardeners kneeling pad keep cal... | 546                        | 2.74                      |  |
| 17                                 | jumbo bag apples                  | jumbo bag red retrospot            | 546                        | 2.74                      |  |
| 18                                 | jumbo bag red retrospot           | jumbo bag red retrospot            | 544                        | 2.73                      |  |
| Total rows: 20                     |                                   | Query complete 00:07:45.340        |                            |                           |  |

| Bagian           | Fungsi                                                           |
|------------------|------------------------------------------------------------------|
| transaksi_produk | Menyiapkan daftar produk per transaksi (invoice)                 |
| kombinasi_produk | Membentuk pasangan produk yang dibeli bersama                    |
| total_transaksi  | Menghitung total transaksi unik                                  |
| Final SELECT     | Menghitung support (% transaksi yang mengandung pasangan produk) |

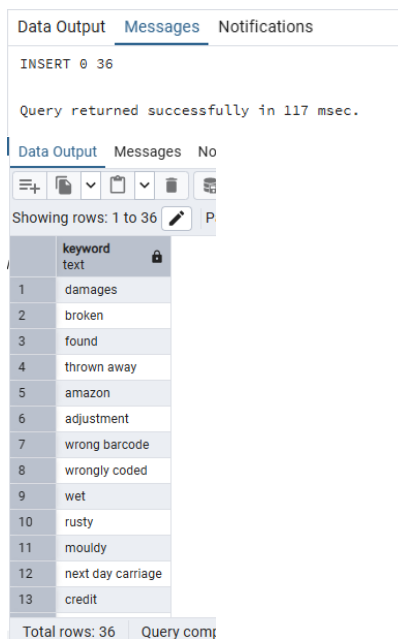
## Analisis Dengan Data Pisah (Data Bersih)

### A. Cleaning Data

#### 1. Pilih Kata Yang Mau Dipisahkan

```
CREATE TEMP TABLE Data_Kotor (keyword TEXT);
```

```
INSERT INTO Data_Kotor(keyword) VALUES  
('damages'), ('broken'), ('found'), ('thrown away'),  
('amazon'), ('adjustment'), ('wrong barcode'),  
('wrongly coded'), ('wet'), ('rusty'), ('mouldy'),  
('next day carriage'), ('credit'), ('crushed'),  
('unsaleable'), ('destroyed'), ('mailout'), ('mixed up'),  
('discount'), ('manual'), ('?'), ('??'), ('???'),  
('?missing'), ('????missing'), ('? sold as sets?'), ('???lost'),  
('B'), ('C2'), ('CRUK'), ('D'), ('DOT'), ('M'), ('PADS'), ('POST'), ('S');
```



The screenshot shows a database interface with tabs for 'Data Output', 'Messages', and 'Notifications'. The 'Messages' tab is active, displaying the message: 'INSERT 0 36' and 'Query returned successfully in 117 msec.'. Below this, the 'Data Output' tab is active, showing a table with 36 rows. The table has a single column named 'keyword' with a data type of 'text'. The first 13 rows of the table are visible, containing the following keywords: 'damages', 'broken', 'found', 'thrown away', 'amazon', 'adjustment', 'wrong barcode', 'wrongly coded', 'wet', 'rusty', 'mouldy', 'next day carriage', and 'credit'. The table footer indicates 'Total rows: 36' and 'Query completed'.

|    | keyword<br>text   |
|----|-------------------|
| 1  | damages           |
| 2  | broken            |
| 3  | found             |
| 4  | thrown away       |
| 5  | amazon            |
| 6  | adjustment        |
| 7  | wrong barcode     |
| 8  | wrongly coded     |
| 9  | wet               |
| 10 | rusty             |
| 11 | mouldy            |
| 12 | next day carriage |
| 13 | credit            |

#### 2. Buat Tabel Data Kotor untuk menyimpan data kotor yang berisi keyword yang di buat

```
CREATE TABLE dirty_data AS  
SELECT *  
FROM online_retail o  
WHERE EXISTS (  
    SELECT 1  
    FROM Data_Kotor dk  
    WHERE LOWER(o.stockcode) = LOWER(dk.keyword)  
);
```

| Data Output Messages Notifications                             |                   |                   |                     |                     |                     |                   |                       |  |
|----------------------------------------------------------------|-------------------|-------------------|---------------------|---------------------|---------------------|-------------------|-----------------------|--|
| Showing rows: 1 to 1000 of 3 Page No: 1                        |                   |                   |                     |                     |                     |                   |                       |  |
|                                                                | invoiceno<br>text | stockcode<br>text | description<br>text | quantity<br>integer | invoicedate<br>text | unitprice<br>text | customerid<br>integer |  |
| 1                                                              | 536370            | POST              | POSTAGE             | 3                   | 01/12/10 08.45      | 18                | 12583                 |  |
| 2                                                              | C536379           | D                 | Discount            | -1                  | 01/12/10 09.41      | 27,5              | 14527                 |  |
| 3                                                              | 536403            | POST              | POSTAGE             | 1                   | 01/12/10 11.27      | 15                | 12791                 |  |
| 4                                                              | 536527            | POST              | POSTAGE             | 1                   | 01/12/10 13.04      | 18                | 12662                 |  |
| 5                                                              | 536540            | C2                | CARRIAGE            | 1                   | 01/12/10 14.05      | 50                | 14911                 |  |
| 6                                                              | 536544            | DOT               | DOTCOM POSTAGE      | 1                   | 01/12/10 14.32      | 569,77            | [null]                |  |
| 7                                                              | 536569            | M                 | Manual              | 1                   | 01/12/10 15.35      | 1,25              | 16274                 |  |
| 8                                                              | 536569            | M                 | Manual              | 1                   | 01/12/10 15.35      | 18,95             | 16274                 |  |
| 9                                                              | 536592            | DOT               | DOTCOM POSTAGE      | 1                   | 01/12/10 17.06      | 607,49            | [null]                |  |
| 10                                                             | 536840            | POST              | POSTAGE             | 1                   | 02/12/10 18.27      | 18                | 12736                 |  |
| 11                                                             | 536852            | POST              | POSTAGE             | 1                   | 03/12/10 09.51      | 18                | 12686                 |  |
| Total rows: 2845 Query complete 00:00:00.103 CRLF Ln 1, Col 24 |                   |                   |                     |                     |                     |                   |                       |  |

### 3. Buat Tabel Data Bersih Untuk Mempertahankan Data Utuh/Bersih

```

CREATE TABLE clean_data AS
SELECT *
FROM online_retail o
WHERE NOT EXISTS (
    SELECT 1
    FROM Data_Kotor dk
    WHERE LOWER(o.stockcode) = LOWER(dk.keyword)
);

```

| Data Output Messages Notifications                               |                   |                   |                                      |                     |                     |                   |                       |  |
|------------------------------------------------------------------|-------------------|-------------------|--------------------------------------|---------------------|---------------------|-------------------|-----------------------|--|
| Showing rows: 1 to 1000 of 540 Page No: 1                        |                   |                   |                                      |                     |                     |                   |                       |  |
|                                                                  | invoiceno<br>text | stockcode<br>text | description<br>text                  | quantity<br>integer | invoicedate<br>text | unitprice<br>text | customerid<br>integer |  |
| 1                                                                | 536365            | 85123A            | WHITE HANGING HEART T-LIGHT HOLDER   | 6                   | 01/12/10 08.26      | 2,55              |                       |  |
| 2                                                                | 536365            | 71053             | WHITE METAL LANTERN                  | 6                   | 01/12/10 08.26      | 3,39              |                       |  |
| 3                                                                | 536365            | 84406B            | CREAM CUPID HEARTS COAT HANGER       | 8                   | 01/12/10 08.26      | 2,75              |                       |  |
| 4                                                                | 536365            | 84029G            | KNITTED UNION FLAG HOT WATER BOTT... | 6                   | 01/12/10 08.26      | 3,39              |                       |  |
| 5                                                                | 536365            | 84029E            | RED WOOLLY HOTTIE WHITE HEART.       | 6                   | 01/12/10 08.26      | 3,39              |                       |  |
| 6                                                                | 536365            | 22752             | SET 7 BABUSHKA NESTING BOXES         | 2                   | 01/12/10 08.26      | 7,65              |                       |  |
| 7                                                                | 536365            | 21730             | GLASS STAR FROSTED T-LIGHT HOLDER    | 6                   | 01/12/10 08.26      | 4,25              |                       |  |
| 8                                                                | 536366            | 22633             | HAND WARMER UNION JACK               | 6                   | 01/12/10 08.28      | 1,85              |                       |  |
| 9                                                                | 536366            | 22632             | HAND WARMER RED POLKA DOT            | 6                   | 01/12/10 08.28      | 1,85              |                       |  |
| 10                                                               | 536367            | 84879             | ASSORTED COLOUR BIRD ORNAMENT        | 32                  | 01/12/10 08.34      | 1,69              |                       |  |
| 11                                                               | 536367            | 22745             | POPPY'S PLAYHOUSE BEDROOM            | 6                   | 01/12/10 08.34      | 2,1               |                       |  |
| Total rows: 539064 Query complete 00:00:00.783 CRLF Ln 1, Col 24 |                   |                   |                                      |                     |                     |                   |                       |  |

## B. Explorasi Data

### 1. Net Revenue

```

SELECT
SUM(CASE
    WHEN REPLACE(unitprice, ',', ':')::NUMERIC > 0
    THEN quantity * REPLACE(unitprice, ',', ':')::NUMERIC
    ELSE 0
END) AS total_revenue_positif,
SUM(CASE
    WHEN REPLACE(unitprice, ',', ':')::NUMERIC < 0
    THEN quantity * REPLACE(unitprice, ',', ':')::NUMERIC
    ELSE 0

```

```

END) AS total_return,
SUM(quantity * REPLACE(unitprice, ',', '::NUMERIC) AS revenue_bersih
FROM clean_data;

```

|   | total_revenue_positif<br>numeric | total_return<br>numeric | revenue_bersih<br>numeric |
|---|----------------------------------|-------------------------|---------------------------|
| 1 | 9564698.551                      | 0                       | 9564698.551               |

## 2. Total Revenue

```

SELECT

```

```

COUNT(*) AS total_transaksi,
SUM(quantity * REPLACE(unitprice, ',', '::NUMERIC) AS total_revenue
FROM clean_data;

```

|   | total_transaksi<br>bigint | total_revenue<br>numeric |
|---|---------------------------|--------------------------|
| 1 | 539064                    | 9564698.551              |

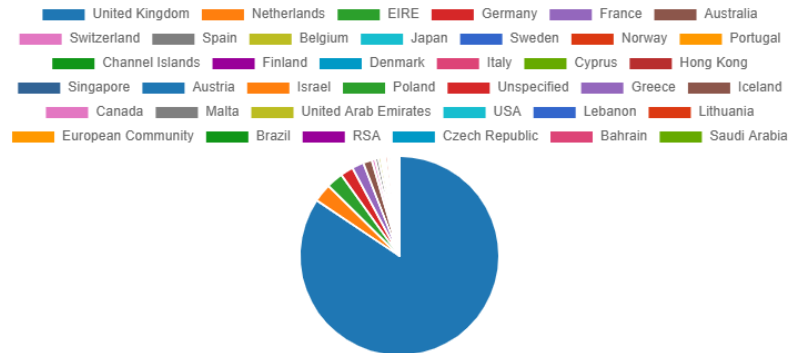
## 3. Total transaksi dan revenue per Negara

```

SELECT country,
COUNT(*) AS total_transaksi,
SUM(quantity::NUMERIC * REPLACE(unitprice, ',', '::NUMERIC) AS total_revenue
FROM clean_data
GROUP BY country
ORDER BY total_revenue DESC;

```

| Data Output Messages Notifications    |                 |                             |                          |
|---------------------------------------|-----------------|-----------------------------|--------------------------|
| Showing rows: 1 to 38 Page No: 1 of 1 |                 |                             |                          |
|                                       | country<br>text | total_transaksi<br>bigint   | total_revenue<br>numeric |
| 1                                     | United Kingdom  | 493954                      | 8073707.951              |
| 2                                     | Netherlands     | 2330                        | 283479.54                |
| 3                                     | EIRE            | 8075                        | 259663.46                |
| 4                                     | Germany         | 9096                        | 200808.40                |
| 5                                     | France          | 8234                        | 182163.39                |
| 6                                     | Australia       | 1257                        | 136990.00                |
| 7                                     | Switzerland     | 1969                        | 52505.35                 |
| 8                                     | Spain           | 2468                        | 51765.20                 |
| 9                                     | Belgium         | 1971                        | 36662.96                 |
| 10                                    | Japan           | 355                         | 35419.79                 |
| 11                                    | Sweden          | 437                         | 35176.91                 |
| 12                                    | Norway          | 1060                        | 32292.96                 |
| 13                                    | Portugal        | 1475                        | 26871.52                 |
| 14                                    | Channel Islands | 753                         | 19936.29                 |
| 15                                    | Finland         | 653                         | 18202.54                 |
| Total rows: 38                        |                 | Query complete 00:00:00.518 |                          |



#### 4. Negara dengan transaksi Terbanyak

SELECT country,

COUNT(\*) AS total\_transaksi

FROM clean\_data

GROUP BY country

ORDER BY total\_transaksi DESC

LIMIT 1;

| Data Output Messages Graph Visuali |                 |                           |
|------------------------------------|-----------------|---------------------------|
| Showing rows: 1 to 1 Page No: 1    |                 |                           |
|                                    | country<br>text | total_transaksi<br>bigint |
| 1                                  | United Kingdom  | 493954                    |

#### 5. Revenue Perbulan

SELECT

TO\_CHAR(TO\_TIMESTAMP(invoicedate, 'DD/MM/YY HH24:MI'), 'YYYY-MM') AS bulan,

SUM(quantity \* REPLACE(unitprice, ',', ':')::NUMERIC) AS total\_revenue

FROM clean\_data

GROUP BY 1

ORDER BY 1;



| Data Output Messages Graph |               |                          |
|----------------------------|---------------|--------------------------|
| Showing rows: 1 to 13      |               |                          |
|                            | bulan<br>text | total_revenue<br>numeric |
| 1                          | 2010-12       | 720110.12                |
| 2                          | 2011-01       | 546089.23                |
| 3                          | 2011-02       | 488782.82                |
| 4                          | 2011-03       | 669288.95                |
| 5                          | 2011-04       | 482532.19                |
| 6                          | 2011-05       | 718849.36                |
| 7                          | 2011-06       | 714243.21                |
| 8                          | 2011-07       | 667085.54                |
| 9                          | 2011-08       | 679414.55                |
| 10                         | 2011-09       | 1000169.301              |
| 11                         | 2011-10       | 1048082.08               |
| 12                         | 2011-11       | 1418084.32               |
| 13                         | 2011-12       | 411966.88                |

## 6. Total Transaksi Dan Revenue per hari

```

SELECT
  TO_CHAR(TO_TIMESTAMP(invoicedate, 'DD/MM/YY HH24:MI'), 'YYYY-MM-DD')
  AS tanggal,
  COUNT(*) AS total_transaksi,
  SUM(quantity * REPLACE(unitprice, ',', '::NUMERIC) AS total_revenue
FROM clean_data
GROUP BY 1
ORDER BY 1;

```

| Data Output Messages Graph Visualiser x Noti |                 |                           |                          |
|----------------------------------------------|-----------------|---------------------------|--------------------------|
| Showing rows: 1 to 305                       |                 |                           |                          |
|                                              | tanggal<br>text | total_transaksi<br>bigint | total_revenue<br>numeric |
| 1                                            | 2010-12-01      | 3099                      | 57328.60                 |
| 2                                            | 2010-12-02      | 2108                      | 46189.28                 |
| 3                                            | 2010-12-03      | 2187                      | 43510.73                 |
| 4                                            | 2010-12-05      | 2712                      | 30957.28                 |
| 5                                            | 2010-12-06      | 3866                      | 50907.56                 |
| 6                                            | 2010-12-07      | 2948                      | 41792.56                 |
| 7                                            | 2010-12-08      | 2639                      | 43375.58                 |
| 8                                            | 2010-12-09      | 2873                      | 48978.46                 |
| 9                                            | 2010-12-10      | 2752                      | 55205.10                 |
| 10                                           | 2010-12-12      | 1451                      | 17240.92                 |
| 11                                           | 2010-12-13      | 2277                      | 35458.84                 |
| 12                                           | 2010-12-14      | 2077                      | 42007.09                 |
| 13                                           | 2010-12-15      | 1342                      | 29517.91                 |
| 14                                           | 2010-12-16      | 1788                      | 48088.35                 |
| 15                                           | 2010-12-17      | 3094                      | 39404.72                 |
| Total rows: 305 Query complete 00:00:00.794  |                 |                           |                          |

## 7. Total Revenue CID

```

SELECT customerid,
  SUM(quantity::NUMERIC * REPLACE(unitprice, ',', '::NUMERIC) AS total_revenue
FROM clean_data
GROUP BY customerid
ORDER BY total_revenue DESC;

```

| Data Output Messages Graph Visuali |                       |                          |
|------------------------------------|-----------------------|--------------------------|
| Showing rows: 1 to 1000 Page No:   |                       |                          |
|                                    | customerid<br>integer | total_revenue<br>numeric |
| 1                                  | [null]                | 1277870.23               |
| 2                                  | 14646                 | 278778.02                |
| 3                                  | 18102                 | 259657.30                |
| 4                                  | 17450                 | 189735.53                |
| 5                                  | 14911                 | 128882.13                |
| 6                                  | 12415                 | 123638.18                |
| 7                                  | 14156                 | 113855.32                |
| 8                                  | 17511                 | 88138.20                 |
| 9                                  | 16684                 | 65920.12                 |
| 10                                 | 13694                 | 62924.10                 |
| 11                                 | 15311                 | 59419.34                 |
| 12                                 | 16029                 | 58193.85                 |
| 13                                 | 13089                 | 57385.88                 |
| 14                                 | 15061                 | 54250.34                 |
| 15                                 | 14096                 | 53258.43                 |
| Total rows: 4364                   |                       | Query complete 00:       |

## 8. Rata Rata Penjualan

SELECT AVG(quantity) AS avg\_quantity  
FROM clean\_data;

| Data Output Messages |                         |
|----------------------|-------------------------|
| Showing rows: 1 to 1 |                         |
|                      | avg_quantity<br>numeric |
| 1                    | 9.5836431295727409      |

## 9. TOP 10 produk

SELECT  
description,  
SUM(quantity) AS total\_terjual  
FROM clean\_data  
GROUP BY description  
ORDER BY total\_terjual DESC  
LIMIT 10;

| Data Output Messages Graph Visualiser SQL |                                    |                             |
|-------------------------------------------|------------------------------------|-----------------------------|
| Showing rows: 1 to 10 Page No: 1 of 1     |                                    |                             |
|                                           | description<br>text                | total_terjual<br>bigint     |
| 1                                         | WORLD WAR 2 GLIDERS ASSTD DESIG... | 53847                       |
| 2                                         | JUMBO BAG RED RETROSPOT            | 47363                       |
| 3                                         | ASSORTED COLOUR BIRD ORNAMENT      | 36381                       |
| 4                                         | POPCORN HOLDER                     | 36334                       |
| 5                                         | PACK OF 72 RETROSPOT CAKE CASES    | 36039                       |
| 6                                         | WHITE HANGING HEART T-LIGHT HOL... | 35317                       |
| 7                                         | RABBIT NIGHT LIGHT                 | 30680                       |
| 8                                         | MINI PAINT SET VINTAGE             | 26437                       |
| 9                                         | PACK OF 12 LONDON TISSUES          | 26315                       |
| 10                                        | PACK OF 60 PINK PAISLEY CAKE CASES | 24753                       |
| Total rows: 10                            |                                    | Query complete 00:00:00.203 |

## 10. Top Produk Menguntungkan Berdasarkan Profit

```
SELECT description,  
       SUM(quantity::NUMERIC * REPLACE(unitprice, ',', ':')::NUMERIC) AS total_profit  
FROM online_retail  
GROUP BY description  
HAVING SUM(quantity::NUMERIC * REPLACE(unitprice, ',', ':')::NUMERIC) > 0  
ORDER BY total_profit DESC  
LIMIT 10;
```

| Data Output Messages Graph Visualiser X Notifications |                                    |                         |  |
|-------------------------------------------------------|------------------------------------|-------------------------|--|
| Showing rows: 1 to 10 Page No: 1 of 1                 |                                    |                         |  |
|                                                       | description<br>text                | total_profit<br>numeric |  |
| 1                                                     | REGENCY CAKESTAND 3 TIER           | 164762.19               |  |
| 2                                                     | WHITE HANGING HEART T-LIGHT HOLDER | 99668.47                |  |
| 3                                                     | PARTY BUNTING                      | 98302.98                |  |
| 4                                                     | JUMBO BAG RED RETROSPOT            | 92356.03                |  |
| 5                                                     | RABBIT NIGHT LIGHT                 | 66756.59                |  |
| 6                                                     | PAPER CHAIN KIT 50'S CHRISTMAS     | 63791.94                |  |
| 7                                                     | ASSORTED COLOUR BIRD ORNAMENT      | 58959.73                |  |
| 8                                                     | CHILLI LIGHTS                      | 53768.06                |  |
| 9                                                     | SPOTTY BUNTING                     | 42065.32                |  |
| 10                                                    | JUMBO BAG PINK POLKADOT            | 41619.66                |  |

## 11. Bottom 10 Produk (Penjualan)

```
SELECT  
description,  
SUM(quantity) AS total_terjual  
FROM clean_online_retail  
GROUP BY description  
HAVING SUM(quantity) > 0  
ORDER BY total_terjual ASC  
LIMIT 10;
```

| Data Output Messages Graph Visualiser X Notifications |                                     |                         |  |
|-------------------------------------------------------|-------------------------------------|-------------------------|--|
| Showing rows: 1 to 10 Page No: 1 of 1                 |                                     |                         |  |
|                                                       | description<br>text                 | total_terjual<br>bigint |  |
| 1                                                     | PINK FEATHER CHRISTMAS DECORATION   | 1                       |  |
| 2                                                     | Amazon Adjustment                   | 1                       |  |
| 3                                                     | SET 10 CARDS CHRISTMAS BAUBLE 16954 | 1                       |  |
| 4                                                     | SET 10 CARDS SNOWY ROBIN 17099      | 1                       |  |
| 5                                                     | SET 10 CARDS 3 WISE MEN 17107       | 1                       |  |
| 6                                                     | BLUE CRYSTAL BOOT PHONE CHARM       | 1                       |  |
| 7                                                     | SET 36 COLOURING PENCILS DOILEY     | 1                       |  |
| 8                                                     | GREEN PENDANT SHELL NECKLACE        | 1                       |  |
| 9                                                     | JARDIN ETCHED GLASS BUTTER DISH     | 1                       |  |
| 10                                                    | BLUE NEW BAROQUE FLOCK CANDLESTL... | 1                       |  |

## 12. Bottom 10 Produk (Revenue)

```
SELECT description,  
       SUM(quantity::NUMERIC * REPLACE(unitprice, ',', ':')::NUMERIC) AS total_revenue
```

```

FROM clean_data
GROUP BY description
HAVING SUM(quantity::NUMERIC * REPLACE(unitprice, ',', '. '::NUMERIC) < 0
ORDER BY total_revenue ASC
LIMIT 10;

```

| Data Output Messages Graph Visualiser x Notificatio |                                 |                          |
|-----------------------------------------------------|---------------------------------|--------------------------|
| Showing rows: 1 to 10 Page No: 1 of 1               |                                 |                          |
|                                                     | description<br>text             | total_revenue<br>numeric |
| 1                                                   | AMAZON FEE                      | -221520.50               |
| 2                                                   | Bank Charges                    | -7175.639                |
| 3                                                   | WHITE CHERRY LIGHTS             | -54.00                   |
| 4                                                   | CREAM SWEETHEART MAGAZINE R...  | -46.85                   |
| 5                                                   | WOODEN BOX ADVENT CALENDAR      | -45.70                   |
| 6                                                   | ASSORTED TUTTI FRUTTI ROUND BOX | -39.60                   |
| 7                                                   | PINK CHERRY LIGHTS              | -27.00                   |
| 8                                                   | BLUE PADDED SOFT MOBILE         | -25.50                   |
| 9                                                   | ANTIQUE LILY FAIRY LIGHTS       | -14.85                   |
| 10                                                  | CREAM SWEETHEART TRAYS          | -12.75                   |

## C. Analisis Prediksi

### 1. Transaksi Tanpa CID/Anonim

```

SELECT
COUNT(*) AS transaksi_anonim,
SUM(quantity::NUMERIC * REPLACE(unitprice, ',', '. '::NUMERIC) AS
nilai_transaksi_anonim
FROM clean_data
WHERE customerid IS NULL;

```

| Data Output Messages Graph Visualiser x Not |                            |                                   |
|---------------------------------------------|----------------------------|-----------------------------------|
| Showing rows: 1 to 1 Page No: 1             |                            |                                   |
|                                             | transaksi_anonim<br>bigint | nilai_transaksi_anonim<br>numeric |
| 1                                           | 134143                     | 1277870.23                        |

### 2. Linear Regression Manual

```

WITH penjualan_bulanan AS (
SELECT
DATE_TRUNC('month', TO_TIMESTAMP(invoicedate, 'DD/MM/YY HH24:MI')) AS
bulan,
SUM(quantity::NUMERIC * REPLACE(unitprice, ',', '. '::NUMERIC) AS total_penjualan
FROM clean_data
WHERE quantity::NUMERIC > 0 AND REPLACE(unitprice, ',', '. '::NUMERIC > 0
GROUP BY 1
),
data_berlabel AS (
SELECT

```

```

    ROW_NUMBER() OVER (ORDER BY bulan) AS bulan_ke,
    bulan,
    total_penjualan
FROM penjualan_bulanan
)
SELECT * FROM data_berlabel;

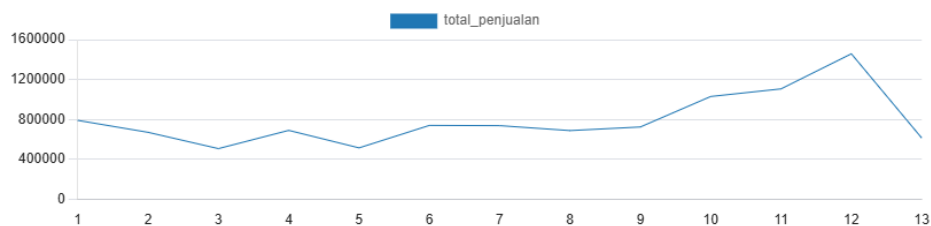
```

Data Output Messages Graph Visualiser x Notifications

Showing rows: 1 to 13 Page No: 1 of 1

|    | bulan_ke<br>bigint | bulan<br>timestamp with time zone | total_penjualan<br>numeric |
|----|--------------------|-----------------------------------|----------------------------|
| 1  | 1                  | 2010-12-01 00:00:00+07            | 791564.69                  |
| 2  | 2                  | 2011-01-01 00:00:00+07            | 672007.41                  |
| 3  | 3                  | 2011-02-01 00:00:00+07            | 508952.87                  |
| 4  | 4                  | 2011-03-01 00:00:00+07            | 691485.70                  |
| 5  | 5                  | 2011-04-01 00:00:00+07            | 516324.79                  |
| 6  | 6                  | 2011-05-01 00:00:00+07            | 741290.74                  |
| 7  | 7                  | 2011-06-01 00:00:00+07            | 738876.88                  |
| 8  | 8                  | 2011-07-01 00:00:00+07            | 689397.74                  |
| 9  | 9                  | 2011-08-01 00:00:00+07            | 725605.16                  |
| 10 | 10                 | 2011-09-01 00:00:00+07            | 1030530.361                |
| 11 | 11                 | 2011-10-01 00:00:00+07            | 1106716.49                 |
| 12 | 12                 | 2011-11-01 00:00:00+07            | 1457775.53                 |
| 13 | 13                 | 2011-12-01 00:00:00+07            | 615516.60                  |

Total rows: 13 Query complete 00:00:01.475



### 3. Hitung Slope dan Intercept (Trendline Regression)

```

WITH data_berlabel AS (
    SELECT
        ROW_NUMBER() OVER (ORDER BY DATE_TRUNC('month',
        TO_TIMESTAMP(invoicedate, 'DD/MM/YY HH24:MI')))) AS bulan_ke,
        DATE_TRUNC('month', TO_TIMESTAMP(invoicedate, 'DD/MM/YY HH24:MI')) AS
        bulan,
        SUM(quantity::NUMERIC * REPLACE(unitprice, ',', '::NUMERIC) AS total_penjualan
    FROM clean_data
    WHERE quantity::NUMERIC > 0 AND REPLACE(unitprice, ',', '::NUMERIC) > 0
    GROUP BY DATE_TRUNC('month', TO_TIMESTAMP(invoicedate, 'DD/MM/YY
    HH24:MI'))
),
statistik AS (
    SELECT
        COUNT(*) AS n,
        SUM(bulan_ke) AS sum_x,

```

```

SUM(total_penjualan) AS sum_y,
SUM(bulan_ke * total_penjualan) AS sum_xy,
SUM(bulan_ke * bulan_ke) AS sum_xx
FROM data_berlabel
),
regresi AS (
SELECT
(n * sum_xy - sum_x * sum_y) * 1.0 / (n * sum_xx - sum_x * sum_x) AS slope,
(sum_y - ((n * sum_xy - sum_x * sum_y) * 1.0 / (n * sum_xx - sum_x * sum_x)) * sum_x)
/ n AS intercept
FROM statistik
)
SELECT
d.bulan_ke,
d.bulan,
d.total_penjualan,
ROUND(r.slope * (d.bulan_ke + 1) + r.intercept, 2) AS prediksi_bulan_depan,
ROUND(r.slope * (d.bulan_ke + 2) + r.intercept, 2) AS prediksi_2_bulan_lagi
FROM data_berlabel d
CROSS JOIN regresi r
ORDER BY d.bulan_ke DESC
LIMIT 1;

```

| Data Output Messages Graph Visualiser x Notifications |                                   |                            |                                 |                                  |  |
|-------------------------------------------------------|-----------------------------------|----------------------------|---------------------------------|----------------------------------|--|
| Showing rows: 1 to 1 Page No: 1 of 1                  |                                   |                            |                                 |                                  |  |
| bulan_ke<br>bigint                                    | bulan<br>timestamp with time zone | total_penjualan<br>numeric | prediksi_bulan_depan<br>numeric | prediksi_2_bulan_lagi<br>numeric |  |
| 13                                                    | 2011-12-01 00:00:00+07            | 615516.60                  | 1046903.78                      | 1083428.00                       |  |

### Hasil:

- Dapat prediksi bulan depan dan dua bulan ke depan berdasarkan linear trend dari historis penjualan.
- Tidak sepresisi machine learning, tapi cukup kuat sebagai dasar manajemen inventory dan perencanaan.

| Temuan         | Implikasi Bisnis                                |
|----------------|-------------------------------------------------|
| Tren meningkat | Siapkan stok tambahan, perkuat kampanye iklan   |
| Tren menurun   | Evaluasi harga/promosi, aktifkan pelanggan lama |
| Tren stagnan   | Eksplorasi diversifikasi produk atau channel    |

## D. ANALISIS MBA

### 1. Data transaksi yang membeli lebih dari 1 produk

```
SELECT invoiceno, LOWER(TRIM(description)) AS item
FROM clean_data
WHERE quantity > 0;
```

|                    | invoiceno<br>text | item<br>text                        |
|--------------------|-------------------|-------------------------------------|
| 1                  | 536365            | white hanging heart t-light holder  |
| 2                  | 536365            | white metal lantern                 |
| 3                  | 536365            | cream cupid hearts coat hanger      |
| 4                  | 536365            | knitted union flag hot water bottle |
| 5                  | 536365            | red woolly hottie white heart.      |
| 6                  | 536365            | set 7 babushka nesting boxes        |
| 7                  | 536365            | glass star frosted t-light holder   |
| 8                  | 536366            | hand warmer union jack              |
| 9                  | 536366            | hand warmer red polka dot           |
| 10                 | 536367            | assorted colour bird ornament       |
| 11                 | 536367            | poppy's playhouse bedroom           |
| 12                 | 536367            | poppy's playhouse kitchen           |
| 13                 | 536367            | feltcraft princess charlotte doll   |
| 14                 | 536367            | ivory knitted mug cosy              |
| 15                 | 536367            | box of 6 assorted colour teaspoon   |
| Total rows: 528967 |                   | Query complete 00:00:01.293         |

### 2. Buat tabel transaksi

```
CREATE TEMP TABLE transaksi AS
SELECT DISTINCT invoiceno, LOWER(TRIM(description)) AS item
FROM clean_data
WHERE quantity > 0;
```

|                    | invoiceno<br>text | item<br>text                        |
|--------------------|-------------------|-------------------------------------|
| 1                  | 540255            | blue charlie+lola personal doorsign |
| 2                  | 539483            | grey floral feltcraft shoulder bag  |
| 3                  | 537395            | man flu metal sign                  |
| 4                  | 540404            | lunch bag red retro spot            |
| 5                  | 537240            | magic drawing slate purdey          |
| 6                  | 540704            | birdhouse garden marker             |
| 7                  | 539629            | queens guard coffee mug             |
| 8                  | 540418            | small heart measuring spoons        |
| 9                  | 538877            | woodland charlotte bag              |
| 10                 | 542192            | red retro spot wrap                 |
| 11                 | 538153            | doormat multicolour stripe          |
| 12                 | 541018            | doormat welcome puppies             |
| Total rows: 518172 |                   | Query complete 00:00:00.270         |

**Hasil :** dalam 1 invoiceno ada produk yang sama muncul lebih dari sekali, itu akan dihapus duplikatnya. Sehingga Tidak perlu tahu **berapa banyak** produk dibeli, cukup tahu **produk apa saja**

### 3. Buat pasangan item A & B dalam transaksi yang sama

```
CREATE TEMP TABLE pasangan_item AS
SELECT
  t1.item AS item_a,
  t2.item AS item_b,
  t1.invoiceno
FROM transaksi t1
JOIN transaksi t2
  ON t1.invoiceno = t2.invoiceno
WHERE t1.item < t2.item; -- untuk menghindari duplikat dan A = A
```

| Data Output Messages Graph Visualiser x Notifications |                                     |                                     |                   |
|-------------------------------------------------------|-------------------------------------|-------------------------------------|-------------------|
| Showing rows: 1 to 1000 Page No: 1 of 28334           |                                     |                                     |                   |
|                                                       | item_a<br>text                      | item_b<br>text                      | invoiceno<br>text |
| 1                                                     | glass star frosted t-light holder   | white metal lantern                 | 536365            |
| 2                                                     | glass star frosted t-light holder   | white hanging heart t-light holder  | 536365            |
| 3                                                     | glass star frosted t-light holder   | set 7 babushka nesting boxes        | 536365            |
| 4                                                     | glass star frosted t-light holder   | knitted union flag hot water bottle | 536365            |
| 5                                                     | glass star frosted t-light holder   | red woolly hottie white heart.      | 536365            |
| 6                                                     | white hanging heart t-light holder  | white metal lantern                 | 536365            |
| 7                                                     | set 7 babushka nesting boxes        | white metal lantern                 | 536365            |
| 8                                                     | set 7 babushka nesting boxes        | white hanging heart t-light holder  | 536365            |
| 9                                                     | knitted union flag hot water bottle | white metal lantern                 | 536365            |
| 10                                                    | knitted union flag hot water bottle | white hanging heart t-light holder  | 536365            |
| 11                                                    | knitted union flag hot water bottle | set 7 babushka nesting boxes        | 536365            |
| 12                                                    | knitted union flag hot water bottle | red woolly hottie white heart.      | 536365            |
| 13                                                    | red woolly hottie white heart.      | white metal lantern                 | 536365            |
| 14                                                    | red woolly hottie white heart.      | white hanging heart t-light holder  | 536365            |
| 15                                                    | red woolly hottie white heart       | set 7 babushka nesting boxes        | 536365            |
| Total rows: 28333296                                  |                                     | Query complete 00:00:34.917         |                   |

### 4. Hitung total transaksi dan frekuensi

```
-- Total transaksi
SELECT COUNT(DISTINCT invoiceno) FROM transaksi; -- misalnya hasilnya 19905
```

| Data Output |                 |
|-------------|-----------------|
|             | count<br>bigint |
| 1           | 20554           |

### 5. Hitung support, confidence, lift

```
SELECT
  item_a,
  item_b,
```



COUNT(\*)::DECIMAL AS jumlah\_bersama,

-- Support: kombinasi A dan B muncul berapa kali dibagi total transaksi  
COUNT(\*)::DECIMAL / 20554 AS support,

-- Confidence: seberapa sering B dibeli saat A dibeli

COUNT(\*)::DECIMAL / (  
SELECT COUNT(DISTINCT invoiceno)  
FROM transaksi  
WHERE item = p.item\_a  
) AS confidence\_a\_to\_b,

-- Lift: seberapa besar pengaruh A terhadap pembelian B

(  
COUNT(\*)::DECIMAL / 20554  
) / (  
(  
SELECT COUNT(DISTINCT invoiceno)  
FROM transaksi  
WHERE item = p.item\_a  
:::DECIMAL / 20554  
\*  
(  
SELECT COUNT(DISTINCT invoiceno)  
FROM transaksi  
WHERE item = p.item\_b  
:::DECIMAL / 20554  
) AS lift

FROM pasangan\_item p  
GROUP BY item\_a, item\_b  
HAVING COUNT(\*) >= 10 -- filter support minimal  
ORDER BY support DESC  
LIMIT 20;

| Data Output Messages Graph Visualiser x Notifications |                                 |                                    |                           |                        |                              |                         |  |
|-------------------------------------------------------|---------------------------------|------------------------------------|---------------------------|------------------------|------------------------------|-------------------------|--|
| Showing rows: 1 to 20 Page No: 1 of 1                 |                                 |                                    |                           |                        |                              |                         |  |
|                                                       | item_a<br>text                  | item_b<br>text                     | jumlah_bersama<br>numeric | support<br>numeric     | confidence_a_to_b<br>numeric | lift<br>numeric         |  |
| 1                                                     | jumbo bag pink polkadot         | jumbo bag red retrospot            | 825                       | 0.04013817261846842464 | 0.67733990147783251232       | 6.65489690964405806158  |  |
| 2                                                     | green regency teacup and saucer | roses regency teacup and saucer    | 768                       | 0.03736498978301060621 | 0.75665024630541871921       | 14.58929564968252940551 |  |
| 3                                                     | jumbo bag red retrospot         | jumbo storage bag suki             | 724                       | 0.03522428724335895689 | 0.34608030592734225621       | 6.00788395948529791857  |  |
| 4                                                     | jumbo bag red retrospot         | jumbo shopper vintage red paisl... | 680                       | 0.03308358470370730758 | 0.32504780114722753346       | 5.68598511045116146169  |  |
| 5                                                     | lunch bag black skull.          | lunch bag red retrospot            | 641                       | 0.03118614381628880023 | 0.50353495679497250589       | 6.61742807030937652662  |  |
| 6                                                     | alarm clock bakelike green      | alarm clock bakelike red           | 640                       | 0.03113749148584217184 | 0.65306122448979591837       | 12.77166546923241229095 |  |
| 7                                                     | green regency teacup and saucer | pink regency teacup and saucer     | 633                       | 0.03079692517271577309 | 0.62364532019704433498       | 16.73421137249353687033 |  |
| 8                                                     | lunch bag pink polkadot         | lunch bag red retrospot            | 606                       | 0.02948331225065680646 | 0.55596330275229357798       | 7.30643844294798094207  |  |
| 9                                                     | pink regency teacup and saucer  | roses regency teacup and saucer    | 599                       | 0.02914274593753040771 | 0.78198433420365535248       | 15.07777298801306955076 |  |
| 10                                                    | jumbo bag baroque black white   | jumbo bag red retrospot            | 585                       | 0.02846161331127761020 | 0.62700964630225080386       | 6.16039974670003012696  |  |
| 11                                                    | jumbo bag red retrospot         | lunch bag red retrospot            | 579                       | 0.02816969932859783984 | 0.27676864244741873805       | 3.63727792638378820056  |  |
| 12                                                    | lunch bag red retrospot         | lunch bag spaceboy design          | 560                       | 0.02724530505011190036 | 0.35805626598465473146       | 6.36083707091494670951  |  |
| 13                                                    | lunch bag red retrospot         | lunch bag suki design              | 556                       | 0.02705069572832538679 | 0.35549872122762148338       | 6.73448913927422301436  |  |
| 14                                                    | lunch bag cars blue             | lunch bag red retrospot            | 554                       | 0.02695339106743213000 | 0.48173913043478260870       | 6.33098854664739241919  |  |
| 15                                                    | paper chain kit 50's christmas  | paper chain kit vintage christmas  | 552                       | 0.02685608640653887321 | 0.47586206896551724138       | 11.92788898233809921879 |  |
| 16                                                    | jumbo bag apples                | jumbo bag red retrospot            | 546                       | 0.02656417242385910285 | 0.55828220858895705521       | 5.48514938591655034347  |  |
| Total rows: 20 Query complete 00:01:00.375            |                                 |                                    | CRLF Ln 38, Col 1         |                        |                              |                         |  |

## E. RFM Analysis (Recency, Frequency, Monetary)

### Tujuan:

Mengukur kualitas dan nilai pelanggan berdasarkan:

- **Recency:** Terakhir belanja
- **Frequency:** Seberapa sering belanja
- **Monetary:** Total nilai belanja

### 1. Buat tabel

```
CREATE TEMP TABLE rfm_raw AS
SELECT
  customerid,
  MAX(TO_DATE(invoicedate, 'DD/MM/YY HH24.MI')) AS last_purchase,
  COUNT(DISTINCT invoiceno) AS frequency,
  SUM(quantity * REPLACE(unitprice, ',', ':')::NUMERIC) AS monetary
FROM clean_data
WHERE quantity > 0 AND customerid IS NOT NULL
GROUP BY customerid;
```

-- Langkah 2: Tambah kolom recency

```
ALTER TABLE rfm_raw ADD COLUMN recency INTEGER;
```

```
UPDATE rfm_raw
SET recency = CURRENT_DATE - last_purchase;
```

| Data Output Messages Graph Visualiser x Notifications |                       |                             |                     |                     |                    |
|-------------------------------------------------------|-----------------------|-----------------------------|---------------------|---------------------|--------------------|
| Showing rows: 1 to 1000 Page No: 1 of 5               |                       |                             |                     |                     |                    |
|                                                       | customerid<br>integer | last_purchase<br>date       | frequency<br>bigint | monetary<br>numeric | recency<br>integer |
| 1                                                     | 12346                 | 2011-01-18                  | 1                   | 77183.60            | 5296               |
| 2                                                     | 12347                 | 2011-12-07                  | 7                   | 4310.00             | 4973               |
| 3                                                     | 12348                 | 2011-09-25                  | 4                   | 1437.24             | 5046               |
| 4                                                     | 12349                 | 2011-11-21                  | 1                   | 1457.55             | 4989               |
| 5                                                     | 12350                 | 2011-02-02                  | 1                   | 294.40              | 5281               |
| 6                                                     | 12352                 | 2011-11-03                  | 7                   | 1385.74             | 5007               |
| 7                                                     | 12353                 | 2011-05-19                  | 1                   | 89.00               | 5175               |
| 8                                                     | 12354                 | 2011-04-21                  | 1                   | 1079.40             | 5203               |
| 9                                                     | 12355                 | 2011-05-09                  | 1                   | 459.40              | 5185               |
| 10                                                    | 12356                 | 2011-11-17                  | 3                   | 2487.43             | 4993               |
| 11                                                    | 12357                 | 2011-11-06                  | 1                   | 6207.67             | 5004               |
| 12                                                    | 12358                 | 2011-12-08                  | 2                   | 928.06              | 4972               |
| 13                                                    | 12359                 | 2011-10-13                  | 4                   | 6372.58             | 5028               |
| 14                                                    | 12360                 | 2011-10-18                  | 3                   | 2302.06             | 5023               |
| 15                                                    | 12361                 | 2011-02-25                  | 1                   | 174.90              | 5258               |
| Total rows: 4336                                      |                       | Query complete 00:00:04.449 |                     |                     |                    |

## 2. Pengelompokan -> Hitung Kuartil (25%, 50%, 75%)

-- Monetary

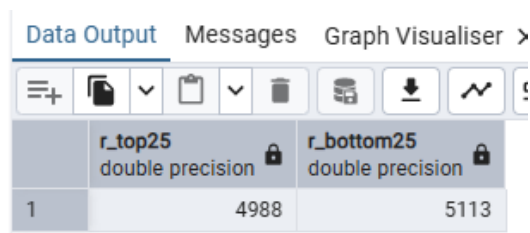
```
SELECT
  PERCENTILE_CONT(0.75) WITHIN GROUP (ORDER BY monetary) AS m_top25,
  PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY monetary) AS m_median,
  PERCENTILE_CONT(0.25) WITHIN GROUP (ORDER BY monetary) AS m_bottom25
FROM rfm_raw;
```

-- Frequency

```
SELECT
  PERCENTILE_CONT(0.75) WITHIN GROUP (ORDER BY frequency) AS f_top25,
  PERCENTILE_CONT(0.25) WITHIN GROUP (ORDER BY frequency) AS f_bottom25
FROM rfm_raw;
```

-- Recency

```
SELECT
  PERCENTILE_CONT(0.25) WITHIN GROUP (ORDER BY recency) AS r_top25,
  PERCENTILE_CONT(0.75) WITHIN GROUP (ORDER BY recency) AS r_bottom25
FROM rfm_raw;
```



|   | r_top25<br>double precision | r_bottom25<br>double precision |
|---|-----------------------------|--------------------------------|
| 1 | 4988                        | 5113                           |

| Metrik    | Top 25%                  | Bottom 25%            |
|-----------|--------------------------|-----------------------|
| Recency   | ≤ 4988 (Aktif)           | ≥ 5113 (Tidak aktif)  |
| Frequency | ≥ 6 (Loyal)              | ≤ 1 (Jarang belanja)  |
| Monetary  | ≥ 2156.35 (High spender) | ≤ 294.4 (Low spender) |

## 3. Klasifikasi Pelanggan Berdasarkan Nilai Kuartil

```
SELECT *,
CASE
  WHEN recency <= 4988 AND frequency >= 6 AND monetary >= 2156.35 THEN
'Champion'
  WHEN recency <= 4988 AND frequency >= 6 THEN 'Loyal Active'
  WHEN recency <= 4988 AND monetary >= 2156.35 THEN 'Potential Big Spender'
  WHEN frequency >= 6 AND monetary >= 2156.35 THEN 'Loyal Big Spender'
```

```

    WHEN recency > 5113 AND frequency <= 1 AND monetary <= 294.4 THEN 'Lost Cheap Buyer'
    WHEN recency > 5113 THEN 'At Risk'
    WHEN frequency <= 1 THEN 'Occasional Buyer'
    WHEN monetary >= 2156.35 THEN 'Big Spender'
    ELSE 'Regular'
END AS segment
FROM rfm_raw;

```

| Data Output Messages Graph Visualiser x Notifications |                       |                       |                     |                     |                    |                  |
|-------------------------------------------------------|-----------------------|-----------------------|---------------------|---------------------|--------------------|------------------|
| Showing rows: 1 to 1000 of 5                          |                       |                       |                     |                     |                    |                  |
|                                                       | customerid<br>integer | last_purchase<br>date | frequency<br>bigint | monetary<br>numeric | recency<br>integer | segment<br>text  |
| 1                                                     | 18119                 | 2010-12-07            | 1                   | 181.12              | 5338               | Lost Cheap Buyer |
| 2                                                     | 18120                 | 2011-05-09            | 1                   | 111.15              | 5185               | Lost Cheap Buyer |
| 3                                                     | 18121                 | 2011-07-13            | 2                   | 673.10              | 5120               | At Risk          |
| 4                                                     | 18122                 | 2011-12-01            | 6                   | 1824.99             | 4979               | Loyal Active     |
| 5                                                     | 18123                 | 2011-11-15            | 2                   | 589.44              | 4995               | Regular          |
| 6                                                     | 18125                 | 2011-11-10            | 4                   | 1142.99             | 5000               | Regular          |
| 7                                                     | 18126                 | 2011-10-10            | 1                   | 204.49              | 5031               | Occasional Buyer |
| 8                                                     | 18127                 | 2011-11-20            | 3                   | 348.73              | 4990               | Regular          |
| 9                                                     | 18129                 | 2011-04-10            | 1                   | 112.55              | 5214               | Lost Cheap Buyer |
| 10                                                    | 18130                 | 2011-11-24            | 3                   | 1059.39             | 4986               | Regular          |
| 11                                                    | 18133                 | 2011-05-11            | 1                   | 931.50              | 5183               | At Risk          |
| 12                                                    | 18135                 | 2011-11-10            | 5                   | 681.91              | 5000               | Regular          |
| 13                                                    | 18136                 | 2011-10-07            | 3                   | 784.33              | 5034               | Regular          |
| 14                                                    | 18138                 | 2011-02-17            | 1                   | 152.65              | 5266               | Lost Cheap Buyer |
| 15                                                    | 18139                 | 2011-11-22            | 6                   | 8438.34             | 4988               | Champion         |
| Total rows: 4336 Query complete 00:00:00.429          |                       |                       |                     |                     |                    |                  |

| Segment                      | Kriteria                                                      |
|------------------------------|---------------------------------------------------------------|
| <b>Champion</b>              | Aktif, sering belanja, nilai tinggi                           |
| <b>Loyal Active</b>          | Aktif dan sering belanja                                      |
| <b>Potential Big Spender</b> | Baru belanja dan nilainya tinggi                              |
| <b>Loyal Big Spender</b>     | Sering belanja dan nilainya tinggi (meskipun sudah agak lama) |
| <b>At Risk</b>               | Sudah lama tidak belanja                                      |
| <b>Lost Cheap Buyer</b>      | Cuma belanja sekali, sudah lama, nilai kecil                  |
| <b>Occasional Buyer</b>      | Jarang belanja ( $\leq 1$ ), tidak melihat waktu atau nilai   |
| <b>Big Spender</b>           | Total belanjanya tinggi, meski tidak sering                   |
| <b>Regular</b>               | Tidak masuk kategori ekstrem lainnya                          |

## F. Prediksi penjualan bulan depan per produk berdasarkan rata-rata 3 bulan terakhir

```

WITH penjualan_bulanan AS (
    SELECT
        description,
        DATE_TRUNC('month', TO_TIMESTAMP(invoicedate, 'DD/MM/YY HH24:MI')) AS
        bulan,
        SUM(quantity) AS total_terjual

```



```

prediksi_penjualan,
ROUND(prediksi_penjualan * 1.5) AS rekomendasi_stok
FROM (
-- ambil dari hasil rata-rata 3 bulan sebelumnya
SELECT
description,
AVG(total_terjual) AS prediksi_penjualan
FROM (
SELECT *,
ROW_NUMBER() OVER (PARTITION BY description ORDER BY bulan DESC)
AS rn
FROM (
SELECT
description,
DATE_TRUNC('month', TO_TIMESTAMP(invoicedate, 'DD/MM/YY HH24:MI')) AS
bulan,
SUM(quantity) AS total_terjual
FROM clean_data
WHERE REPLACE(unitprice, ',', ':')::NUMERIC > 0
GROUP BY description, DATE_TRUNC('month', TO_TIMESTAMP(invoicedate,
'DD/MM/YY HH24:MI'))
) penjualan_bulanan
) recent
WHERE rn <= 3
GROUP BY description
) prediksi;

```

| Data Output Messages Notifications      |                                |                               |                             |
|-----------------------------------------|--------------------------------|-------------------------------|-----------------------------|
| Showing rows: 1 to 1000 Page No: 1 of 5 |                                |                               |                             |
|                                         | description<br>text            | prediksi_penjualan<br>numeric | rekomendasi_stok<br>numeric |
| 1                                       | 4 PURPLE FLOCK DINNER CANDLES  | 33.0000000000000000           | 50                          |
| 2                                       | 50'S CHRISTMAS GIFT BAG LARGE  | 518.6666666666666667          | 778                         |
| 3                                       | DOLLY GIRL BEAKER              | 268.3333333333333333          | 402                         |
| 4                                       | I LOVE LONDON MINI BACKPACK    | 80.6666666666666667           | 121                         |
| 5                                       | I LOVE LONDON MINI RUCKSACK    | 1.0000000000000000            | 2                           |
| 6                                       | NINE DRAWER OFFICE TIDY        | 19.6666666666666667           | 30                          |
| 7                                       | OVAL WALL MIRROR DIAMANTE      | 13.0000000000000000           | 20                          |
| 8                                       | RED SPOT GIFT BAG LARGE        | 424.3333333333333333          | 636                         |
| 9                                       | SET 2 TEA TOWELS I LOVE LONDON | 234.3333333333333333          | 351                         |
| 10                                      | SPACEBOY BABY GIFT SET         | 44.3333333333333333           | 66                          |
| 11                                      | TOADSTOOL BEDSIDE LIGHT        | 2.0000000000000000            | 3                           |
| 12                                      | TRELLIS COAT RACK              | 35.0000000000000000           | 53                          |
| 13                                      | *Boombox Ipod Classic          | 1.0000000000000000            | 2                           |
| 14                                      | *USB Office Mirror Ball        | 2.0000000000000000            | 3                           |
| 15                                      | 10 COI OIUR SPACFROY PFN       | 584.6666666666666667          | 877                         |
| Total rows: 4033                        |                                | Query complete 00:00:02.734   |                             |

## H. Segmentasi Produk Berdasarkan Deskripsi & Penjualan